VEGETATION OF DISTURBED LAND AND WASTE PLACES

Areas which are disturbed by cultivation or some other means and those which may be regarded as waste land do not make up a large part of the Manawatu, but they contain plants and combinations of plants which do not occur in other habitats.

Disturbed land

Disturbance is most commonly brought about by cultivation, trampling, and mass movement of soil by erosion and machinery. Bare ground is a fairly non-selective habitat for pioneering plants. Almost any plant species tolerant of shadeless conditions during establishment may grow there. Most wild alien herbs have the characteristics of pioneers and it is not surprising that they are so much in evidence. Many species of this nature are among the world's most successful plants. Their near-cosmopolitan distribution is a reflection of their adaptability and, to some extent, cause and effect of the wide genetic variability within the species.

Open situations accommodate about 100 common wild plant species. About 40 of these are confined to recently disturbed soils and include many of the most troublesome weeds of cultivated land. Except for *Oxalis latifolia* and *Nothoscordum inodorum*, nearly all are annuals or short-lived perennials. All except *Pelargonium inodorum* are alien plants. Fifteen are summer annuals—3 *Chenopodium* spp., 3 *Amaranthus* spp., 2 *Polygonum* spp., *Galinsoga parviflora, Euphorbia peplus,* summer grass, barnyard grass, *Phalaris minor*, and some forms of *Poa annua.* Some of the variants of *Solanum nigrum* also fall into this group. Small-flowered buttercup, fumitory, *Bromus diandrus, Vulpia* spp., cleavers, subterranean clover, and many other leguminous herbs behave as winter annuals. Spurrey, *Veronica persica,* wild turnip, chickweed, and red dead nettle make most of their growth in winter and spring but have less regular cycles. Some of the obligate occupiers of disturbed ground have their periods of growth regulated more by the periodicity of rainfall than by season. Most frequently they have a flush of growth in the spring and then again at any time when there are significant showers of rain up till the winter. In this group are a few short-lived species (shepherd's purse, groundsel, and bitter cress) and an assortment of plants which are annual to perennial (sow thistle, prickly sow thistle, nipplewort, oxtongue, dove's foot, cut-leaved geranium, and staggerweed). True biennials are relatively few. Hemlock, foxglove, and winter cress come into this group.

Perennial weeds soon assume some importance on disturbed soil. Because of the fairly regular topography of the Manawatu, the generally high fertility

and the moderate rainfall there are few places, except for some sandy areas, where annuals can maintain a permanent hold. Usually short-lived plants and perennials establish together. While the annuals are passing through one or two life spans perennials take hold and increase rapidly in size. Most of this expansion occurs while the annuals are in the seed and seedling stages and offering little competition against the invaders. Among the perennials are pasture grasses and clovers, docks, creeping buttercup, *Veronica serpyllifolia,* ragwort, wireweed, and fennel. Under fertile conditions, and in the absence of further disturbance, grasses and clovers become dominant.

When soil is disturbed the flora which will appear is not very predictable. *Poa annua, Veronica persica,* and chickweed are most likely to be there but the flora depends largely on the history of the piece of ground. On farmland it is not always the weeds of the immediate past which have the greatest bearing on the composition of the new population. Ploughing may bury the superficial seeds and bring to the surface seeds of docks, rushes, plantains, chickweed, the amaranths, thistles, fathen, and shepherd's purse which have lain dormant for a decade or more (*see* Harris 1959). Predictions are complicated further because many weed species have not yet reached all the sites potentially suitable for them. Such plants as the amaranths, *Galinsoga,* and summer grass are still fairly restricted in their distribution.

Waste land

Waste land is regarded here as any piece of land with predominantly non-woody vegetation which is ungrazed, uncultivated, and unmown. Examples of waste land are rubbish dumps, untended places about towns, footpaths, roadsides, and railway margins. Unstable sand country comes within the definition but is treated separately in this handbook because of the distinctive nature of its habitats and communities.

Waste places have considerable botanical interest because of the great diversity of species. However, the vegetation is difficult to describe because the several hundred components co-exist over a wide range of habitats and the different communities are not clearly characterised by a few dominant species. The diversity reflects the wide habitat range, the exposure to sources of new plants, and the enhanced opportunity for establishment in the absence of grazing, cultivation, and mowing. Furthermore, most of the successful new arrivals have a chance of remaining because there is usually no immediate alternative use for the land they occupy.

A good cross section of the district's weeds of open land can be found around rubbish dumps and on untended urban land. A few occur more regularly in these places than elsewhere, particularly *Hirschfeldia incana,* *Raphanus maritimus, Lavatera cretica, Malva neglecta, M. parviflora,* and *Oryzopsis miliacea.*

Cracks in footpaths provide a growing place for almost any weed which happens to find its way there. The most frequent are *Amaranthus deflexus, Poa annua,* sand spurrey, pearlwort, *Cotula australis,* and white clover. *Linaria purpurea* is less common but fairly conspicuous. In some streets in Palmerston North knot-root bristle grass has had a firm hold for many years. *Carex hirta* grew on the unpaved edge of West St, Palmerston North for many years and persisted in cracks after kerbing, channelling, and paving. This is the only known occurrence of the weed in the North Island.

Plant life on the margins of roads and railways has many points of interest. Along roads many habitats are represented because (1) roads traverse nearly every soil type and climatic zone in the district; (2) there are local site differences in the zones from the shingly road margin to the boundary fence; and (3) there are differences caused by discontinuities such as banks and ditches.

The botanical composition of roadside vegetation is influenced by the physical habitat, the stage in vegetational succession, and availability of species. The species which may be found on roadsides fall into three groups: (1) the adaptable species which appear in many diverse situations; (2) those which are almost obligate inhabitants because grazing, cultivation, or mowing exclude them from most other places; and (3) the casuals. Examples are given in the following accounts of roadside vegetation of the sand country, alluvial plains, terraces, downlands, and hills.

Roadsides of the sand country

Sandy roadsides are mostly grassy and have a predominance of species suited to dry conditions. Near the coast the most plentiful grasses are harestail, ratstail, *Bromus diandrus,* and *Poa pratensis.* Some others, less to be expected in this habitat, are Yorkshire fog, perennial ryegrass, and tall fescue. Indian doab is not widespread at present but will become much more prominent. *Notodanthonia* occurs infrequently. Other plants of some significance are sheep's sorrel, *Acaena* spp., the legumes (haresfoot trefoil, black medick, and King Island melilot, together with a group of species which develop from rosettes), *Erigeron floribundus,* catsear, hawkbit, hawksbeard, and *Oenothera stricta.* As well as these are some species which are more widespread on adjoining land—marram, *Scirpus nodosus,* and, in wetter places, *Leptocarpus.*

More stable sand further from the coast has a similar flora but more cocksfoot, *Vulpia* spp., *Aira* spp., tall fescue, and ratstail. In some places knot-root bristle grass is very prominent and apparently increasing. Bracken and *Muehlenbeckia complexa* are of local importance. Around Foxton there

are noticeable amounts of tree mallow, prickly lettuce, Chinese boxthorn, and *Lepidium bonariense.* Of a more casual nature are viper's bugloss, borage, and *Echium lycopsis.*

Roadsides of the alluvial plains

Edges of roads across the plains have an abundance of the regular annual weeds of the district when they are disturbed. In the second phase of succession there is an increasing proportion of perennials. It is at this stage when grasses are rising to prominence that a group of more or less obligate waste land species make an appearance. Prominent among them are the vetches, meadow foxtail, and prairie grass. These are less evident at the cocksfoot and tall fescue stage, which later forms a fairly permanent community. In the moist habitats tall fescue, *Calystegia sepium,* and timothy are more abundant and are replaced in wetter places by *Phalaris arundinacea, Glyceria maxima,* and *Carex lessoniana.* Around Linton *Phalaris aquatica* is quite prominent and competing successfully with tall fescue. It is quite noticeable that rushes have only a minor part in these ungrazed communities. In the first place they have little chance of establishing in the closed communities of tall grasses. Secondly, the rushes are poor competitors with ungrazed grass, *Juncus sarophorus* in particular being unable to hold its place.

There are a number of species confined to this habitat or found here more frequently than elsewhere. Among them are the large-flowered mallow, wild teasel, purple-top, vervain, hops, *Pennisetum macrourum,* and chicory.

The pronounced bands in the roadside vegetation on the alluvial plains can serve as an example of the zonation which occurs on most roadsides to some degree. On the immediate margins many species perpetuate themselves in the shingle which is well watered by run-off from the more impervious metalled or tar-sealed carriageway. *Crassula decumbens,* allseed, sheep's sorrel, *Poa annua,* perennial ryegrass, white clover, browntop, and *Lotus* spp. are frequently encountered but there are wide regional variations. In many places there may be barley, wheat, brassicas, and other farm crops from seeds spilled from passing trucks. Only very few of these plants reach maturity. Other plants such as goat's rue and tree lupin reach roadsides in river gravel which is a rich source of seeds of these and many other weeds. Beyond the shingle is often a band containing stinking mayweed, catchfly, and rayless chamomile if there is fairly regular disturbance, or a strip of grassy vegetation if disturbance is minimal. Where the roadside is used for the passage of farm livestock and implements, ryegrasses and white clover flourish on the more trodden parts, and there may be many weeds of cultivated land if the turf is broken. Near the boundary fence among the tall grasses are the occasional woody plants such as plum, apple, blackberry, Japanese honeysuckle, and *Muehlenbeckia* spp., the latter usually on fences and shelter belts.

Roadsides on the terraces

These have vegetation composed of both native and introduced plants. The succession runs quickly to cocksfoot and tall fescue with appreciable quantities of Yorkshire fog, browntop, and red fescue. On the terraces east of the Manawatu River, in many places undisturbed for 50 years or more, bracken and manuka have been replaced by mahoe and other pre-forest shrubs. Between Palmerston North and Bunnythorpe *Cortaderia toetoe* is quite plentiful and showing no signs of decreasing. The main stands of spiny broom were in this region but many have been reduced or eliminated. Garden refuse has been dumped over banks in many places. It is not unusual to find bamboo, *Tradescantia*, montbretia, *Allium triquetrum*, and *Oxalis incarnata* in these situations. *Rumex sagittatus* has been noted a few times but it has not established.

Roadsides on the downlands

These are mostly grassy but the native flora is well represented by both woody and non-woody plants. The succession to tall grasses is about the same as on the terraces but a surprising feature is the amount of *Lepidosperma australe* which is uncommon elsewhere. Native plants number well over 50. The most noticeable are totara, *Pittosporum tenuifolium, Coprosma* spp., lancewood, *Rubus schmidelioides, Olearia solandri*, manuka, and bracken. At the highest elevations cabbage trees and *Cortaderia toetoe* are plentiful. These communities of native plants are not relics of past vegetation but young communities establishing from seeds and spores from the relics. They have become an attractive feature of the downlands and efforts are being made to preserve them.

Roadsides of the hill country

Only small areas of the hill country are served by roads. Two interesting features are the predominance of short, turfy grasses and the abundance of native plants. The grassy areas have a mat of browntop and red fescue with smaller amounts of *Poa pratensis, Notodanthonia* spp., and *Festuca tenuifolia*, the latter not found elsewhere in the district. The stands of native shrubs have been more difficult to subdue in this region of low fertility, high rainfall, and less intensive agriculture. Prominent among the many shrubs growing there are mahoe, *Fuchsia excorticata*, and pepper tree. In these stands and on exposed roadside banks there are many species of ferns and orchids. The most

plentiful orchids are *Thelymitra longifolia* in several of its forms and *Microtis unifolia*.

Railway lines

Railway tracks have less diversity because the effect of their high artificial embankments overrides, to some extent, the effects of differing soil types and irregularities in topography. Furthermore, railway tracks are less extensive and they also are disturbed less, giving plants of open habitats fewer footholds in a community dominated by cocksfoot and tall fescue. Blackberry, Tangier pea, *Senecio spathulatus*, Oriental mustard, Cape weed, and *Diplotaxis muralis* occur more abundantly on railway verges than elsewhere but nutgrass is possibly the only species found specifically in this habitat. It is worthy of note that the only known occurrence of *Lotus tenuis* in the North Island is beside the road and railway at Milson. This plant, like nutgrass, has potential to grow in other habitats but its slow spread limits its distribution.

Abandonment of the Foxton line and the line through the centre of Palmerston North, followed by the lifting of the rails, produced a temporary community. For a time annual grasses and annual legumes flourished in the loose shingle, with a few other species tolerant of the droughty conditions.

FLORA OF THE MANAWATU

One thousand and thirty-three species, subspecies, varieties, and hybrids are recorded as wild plants in the Manawatu. These are natives of New Zealand persisting as relics of the early vegetation, or exotic plants maintaining themselves without direct help from man. Not all these exotics are truly naturalised as some cannot perpetuate themselves indefinitely in the wild state. These are better defined as casuals.

A total of 505 species, subspecies, varieties, and hybrids, or just under half the wild plants, are indigenous and of these 354 are New Zealand endemics. The indigenous element is greatest among the 85 ferns. All except one are natives and half of these are endemics.

About 350 wild exotic plants of the Manawatu are widespread outside New Zealand. Most are found throughout Europe, Asia, and North Africa. Apart from these Asia alone has contributed 21, North and South America 80, Africa 26, and Australia 14.

One hundred and forty-six families are represented in the Manawatu flora. More than one-quarter of the species are contained in the 3 largest families—Gramineae 109, Compositae 98, Cyperaceae 74. Fewer than one-third of the grasses are indigenous but the sedges are predominantly natives. The high number of species in the Compositae is not unexpected. On a world basis one plant in every 10 is a member of the daisy family. The ratio is about the same in the Manawatu.

The next largest families are very much smaller. Cruciferae has 30 species, Juncaceae 25, Rosaceae 26, and Scrophulariaceae 24. Almost 100 families have fewer than five species in each.

The Manawatu does not have any plants endemic to its area. However, this region may be the only place in New Zealand where *Adiantum formosum* can be found growing in the natural state. *Raoulia tenuicaulis* var. *dimorpha*, *Fuchsia perscandens* and *Libertia peregrinans* were described from plants found in the Manawatu.

The plants listed in this section are arranged alphabetically in families, genera and species within the major groups:

The index (page 191) gives ready access to references to all plants mentioned in this handbook by common or botanical name.

DICOTYLEDONOUS PLANTS

ACANTHACEAE

Acanthus mollis L. Grows wild on the banks of the Manawatu River, in the Terrace End cemetery, on roadsides, and in a few large gardens but is not a weed of importance at present. *S. Europe*

AMARANTHACEAE

Alternanthera denticulata R. Br. Rare. Seen recently only in Palmerston North city on edges of ornamental ponds. A. J. Healy (pers. comm.) recalls seeing it on occasions near Feilding railway station. *N.Z. and other countries*

Amaranthus albus L. "Established in old metal quarry and adjacent waste land, Terrace End, Palmerston North" (Healy 1943). It has disappeared from this locality. I have not seen it in the Manawatu. *N. America*

Amaranthus deflexus L. An abundant summer weed appearing in November and disappearing about June except in sheltered places and under street lamps where it persists throughout the year. It is most abundant in cracks in footpaths, less frequent as a weed in cultivated land although its importance in this habitat is increasing. It alternates seasonally with *Poa annua* in its occupation of sites. *Europe*

Amaranthus lividus L. Only a few plants were noted in the district in 1958 but it is now much more abundant and an important weed in some cropping areas nearby at Opiki. It occurs almost exclusively in disturbed soil and appears to be more abundant where nitrogen levels are high. *Trop. America*

Amaranthus powellii S. Wats. A major weed of cultivated soil but not found in all habitats suited to it. Its enormous seed output will aid its potential for spread. The seeds germinate in November and growth is very rapid until early autumn, when the seeds begin to fall. *N. America*

Amaranthus retroflexus L. A specimen in Botany Division herbarium under this name was collected at Moutoa in 1941 by W. R. Boyce. *Trop. America*

APOCYNACEAE

Parsonsia capsularis (Forst. f.) R. Br. A much less abundant climber than *P. heterophylla*. Although it is widely distributed this species is more frequently seen in drier parts of the Manawatu, particularly in scrub and forest remnants near the coast. *N.Z.*

Parsonsia heterophylla A. Cunn. A climber frequenting most forest remnants. In some places it is quite aggressive but does not destroy the supporting plant as do *Muehlenbeckia* spp. It forms stems in excess of 10 cm in diameter in some forests. *N.Z.*

Vinca major L. Periwinkle. A weed of significance in some semi-shaded places. It was probably brought to the Manawatu as a cover plant for use in large gardens. Periwinkle has been established on running screes in the Manawatu Gorge and is a successful stabiliser. It has replaced other plants in its path as it has spread and shows no sign of allowing any plants to